

CAYLEY, COLLECTED MATHEMATICAL PAPERS, Vol. II.

Scan-verified retype: source PDF pp. 76–104 (book pp. 60–88).

Paper 114 (concluded), Paper 115 (opening).

114 (continued).

**ANALYTICAL RESEARCHES CONNECTED WITH STEINER'S
EXTENSION OF MALFATTI'S PROBLEM.**

[Continuation of § 3, beginning at book p. 60.]

[The opening of § 3, with the surface $2yz + 2zx + 2xy + w^2 = 0$ and the three determinators $-\alpha x + \frac{1}{2\alpha}y + \frac{1}{2\alpha}z + w = 0$, $\frac{1}{2\beta}x - \beta y + \frac{1}{2\beta}z + w = 0$, $\frac{1}{2\gamma}x + \frac{1}{2\gamma}y - \gamma z + w = 0$, was given in the preceding slice. We resume at p. 60, where the tactors are to be found.]

The values of a, b, c remaining to be determined. Now before fixing the values of these quantities, we may find three sections each of them touching a determinator at a point of intersection with the section which corresponds to it of the sections

$$cy - bz = 0, \quad az - cx = 0, \quad bx - ay = 0,$$

and touching the other two of the last-mentioned sections; and when a, b, c have their proper values the sections so found are the *tactors*. For, let $\lambda x + \mu y + \nu z + \rho w = 0$ be the equation of a section touching the determinator $-\alpha x + \frac{1}{2\alpha}y + \frac{1}{2\alpha}z + w = 0$, and the two sections $bx - ay = 0$, $az - cx = 0$; and suppose

$$\Delta^2 = \lambda^2 + \mu^2 + \nu^2 - 2\mu\nu - 2\nu\lambda - 2\lambda\mu - 2\rho^2;$$

the conditions of contact with the sections $bx - ay = 0$, $az - cx = 0$ are found to be

$$\begin{aligned} (b + a) \Delta &= (b + a)\lambda - (b + a)\mu - (b - a)\nu, \\ (c + a) \Delta &= (c + a)\lambda - (c - a)\mu - (c + a)\nu, \end{aligned}$$

values, however, which suppose a correspondence in the signs of the radicals. Thence $(b + a)\mu = (c + a)\nu$; or since the ratios only of the quantities λ, μ, ν, ρ are material, $\mu = c + a$, $\nu = b + a$, and therefore

$$\Delta^2 = \lambda^2 - 2(2a + b + c)\lambda + (b - c)^2 - 2\rho^2 = (\lambda - b - c)^2,$$

or

$$\rho^2 = -2(a\lambda + bc).$$

Hence the equation to a section touching $bx - ay = 0$, $az - cx = 0$ is

$$\lambda x + (c + a)y + (b + a)z + \sqrt{-2(a\lambda + bc)} w = 0,$$

and to express that this touches the determinator in question, we have

$$\pm\alpha(\lambda - b - c) = -\alpha\lambda - \frac{1}{\alpha}(2a + b + c) + 2\sqrt{-2(a\lambda + bc)},$$

and selecting the upper sign,

$$-\lambda - 2a\alpha = -2\sqrt{-2(a\lambda + bc)};$$

whence

$$\lambda = -2a(a\alpha - \sqrt{-2bc}), \quad \sqrt{-2(a\lambda + bc)} = (2a\alpha - \sqrt{-2bc}).$$

Therefore the section touching the determinator and the sections $bx - ay = 0$, $az - cx = 0$ is

$$-2a(a\alpha - \sqrt{-2bc})x + (c + a)y + (b + a)z + (2a\alpha - \sqrt{-2bc})w = 0;$$

and at the point of contact with the determinator

$$2yz + 2zx + 2xy + w^2 = 0.$$

[Book p. 61.] Eliminating w between the first and second equations, and between the second and third equations,

$$\sqrt{-2bc} \left(ax + \frac{1}{2\alpha}y + \frac{1}{2\alpha}z \right) + cy + bz = 0,$$

and from these equations $(cy - bz) = 0$, or the point of contact lies in the section $cy - bz = 0$. It follows that the equations of the *tactors* are

$$\begin{aligned} -2a(a\alpha - \sqrt{-2bc})x + (c + a)y + (b + a)z + (2a\alpha - \sqrt{-2bc})w &= 0, \\ (c + b)x - 2\beta(b\beta - \sqrt{-2ca})y + (a + b)z + (2b\beta - \sqrt{-2ca})w &= 0, \\ (b + c)x + (a + c)y - 2\gamma(c\gamma - \sqrt{-2ab})z + (2c\gamma - \sqrt{-2ab})w &= 0, \end{aligned}$$

where a, b, c still remain to be determined.

Now the separators pass through the point of intersection of the determinators; the equations of these give for the point in question,

$$\begin{aligned} x : y : z : w &= (2\beta\gamma + 1)(-\alpha + \beta + \gamma + 2\alpha\beta\gamma) \\ &: (2\gamma\alpha + 1)(\alpha - \beta + \gamma + 2\alpha\beta\gamma) \\ &: (2\alpha\beta + 1)(\alpha + \beta - \gamma + 2\alpha\beta\gamma) \\ &: -1 + \alpha^2 + \beta^2 + \gamma^2 + 4\alpha^2\beta^2\gamma^2; \end{aligned}$$

and the values of a, b, c are therefore

$$\begin{aligned} a : b : c &= (2\beta\gamma + 1)(-\alpha + \beta + \gamma + 2\alpha\beta\gamma) \\ &: (2\gamma\alpha + 1)(\alpha - \beta + \gamma + 2\alpha\beta\gamma) \\ &: (2\alpha\beta + 1)(\alpha + \beta - \gamma + 2\alpha\beta\gamma), \end{aligned}$$

which are to be substituted for a, b, c in the equations of the separators and tactors respectively.

Now proceeding to find the *bisectors*, let $\lambda x + \mu y + \nu z + \rho w = 0$ be the equation of a section touching the determinators, and suppose as before $\Delta^2 = \lambda^2 + \mu^2 + \nu^2 - 2\mu\nu - 2\nu\lambda - 2\lambda\mu - 2\rho^2$; the conditions of contact are

$$\begin{aligned} \pm\beta\Delta &= \beta\lambda - (\beta + \frac{1}{2\beta})\mu + \beta\nu - 2\rho, \\ \mp\gamma\Delta &= \gamma\lambda + \gamma\mu - (\gamma + \frac{1}{2\gamma})\nu - 2\rho, \end{aligned}$$

[Book p. 62.] where it is necessary, for the present purpose, to give opposite signs to the radicals. For if the radicals had the same sign, it would follow that

$$\frac{1}{\beta} \left[\beta\lambda - (\beta + \frac{1}{2\beta})\mu + \beta\nu - 2\rho \right] - \frac{1}{\gamma} \left[\gamma\lambda + \gamma\mu - (\gamma + \frac{1}{2\gamma})\nu - 2\rho \right] = 0,$$

hence the section $\lambda x + \mu y + \nu z + \rho w = 0$ would pass through the point

$$x : y : z : w = 0 : \frac{1}{2\beta^2} : -\frac{1}{2\gamma^2} : -\frac{1}{\beta} + \frac{1}{\gamma},$$

or the section would be a tangent section of the two determinators of the same class with the resultor $x = 0$, which ought not to be the case. The proper formula is

$$\frac{1}{\beta} \left[\beta\lambda - (\beta + \frac{1}{2\beta})\mu + \beta\nu - 2\rho \right] + \frac{1}{\gamma} \left[\gamma\lambda + \gamma\mu - (\gamma + \frac{1}{2\gamma})\nu - 2\rho \right] = 0,$$

and this equation being satisfied, the section

$$\lambda x + \mu y + \nu z + \rho w = 0$$

passes through a point

$$x : y : z : w = -\frac{1}{\beta^2} - \frac{1}{\gamma^2} : \frac{1}{\gamma^2} : \frac{1}{\beta^2} : -\frac{2}{\beta} - \frac{2}{\gamma}.$$

The bisector passes through this point and the line of intersection of the determinators; its equation is

$$\frac{1}{\beta} \left(\frac{1}{2\beta} x - \beta y + \frac{1}{2\beta} z + w \right) - \frac{1}{\gamma} \left(\frac{1}{2\gamma} x + \frac{1}{2\gamma} y - \gamma z + w \right) = 0,$$

or, reducing and completing the system, the equations of the bisectors are

$$\begin{aligned} \left(\frac{1}{2\beta^2} - \frac{1}{2\gamma^2} \right) x + \left(\gamma + \frac{1}{2\beta^2} \right) y + \left(\frac{1}{2\gamma^2} - \beta \right) z + \left(\frac{1}{\beta} - \frac{1}{\gamma} \right) w &= 0, \\ \left(\frac{1}{2\gamma^2} - \alpha \right) x + \left(\frac{1}{2\gamma^2} - \frac{1}{2\alpha^2} \right) y + \left(\frac{1}{2\alpha^2} - \gamma \right) z + \left(\frac{1}{\gamma} - \frac{1}{\alpha} \right) w &= 0, \\ \left(\alpha - \frac{1}{2\beta^2} \right) x + \left(\frac{1}{2\alpha^2} - \beta \right) y + \left(\frac{1}{2\alpha^2} - \frac{1}{2\beta^2} \right) z + \left(\frac{1}{\alpha} - \frac{1}{\beta} \right) w &= 0. \end{aligned}$$

In order to verify the geometrical construction, it only remains to show that each bisector touches two tactors. Consider the bisector and tactor

$$\begin{aligned} -\left(1 + \frac{c}{2\beta^2} \right) x + \left(1 + \frac{a}{2\gamma^2} \right) y + \left(\frac{a}{2\beta^2} - \frac{b}{2\gamma^2} \right) z - \frac{1}{a} \left(\frac{1}{\gamma} - \frac{1}{\beta} \right) w &= 0, \\ -2a(a\alpha - \sqrt{-2bc})x + (c+a)y + (b+a)z + (2a\alpha - \sqrt{-2bc})w &= 0; \end{aligned}$$

and represent these for a moment by

$$\lambda x + \mu y + \nu z + \rho w = 0, \quad \lambda' x + \mu' y + \nu' z + \rho' w = 0;$$

[Book p. 63.] if Δ be the same as before, and Δ' the like function of $\lambda', \mu', \nu', \rho'$; also if

$$\Phi = \lambda\lambda' + \mu\mu' + \nu\nu' - (\mu\nu' + \mu'\nu) - (\nu\lambda' + \nu'\lambda) - (\lambda\mu' + \lambda'\mu) - 2\rho\rho',$$

then

$$\begin{aligned} \Delta^2 &= \left(2 + \frac{1}{a\beta} \right)^2, \\ \Delta'^2 &= (2a\alpha^2 - 2\alpha\sqrt{-2bc} + b + c)^2, \\ \Phi &= a\alpha^2 \left(2 + \frac{1}{a\beta} \right)^2 - 2\alpha\sqrt{-2bc} \left(2 + \frac{1}{a\beta} \right) + c \left(2 + \frac{1}{a\beta} \right), \end{aligned}$$

and the condition of contact $\Delta\Delta' = \Phi$ (taking the proper sign for the radicals) becomes

$$\left(2 + \frac{1}{a\beta} \right) (2a\alpha^2 - 2\alpha\sqrt{-2bc} + b + c) = a\alpha^2 \left(2 + \frac{1}{a\beta} \right) - 2\alpha\sqrt{-2bc} \left(2 + \frac{1}{a\beta} \right) + c \left(2 + \frac{1}{a\beta} \right),$$

or reducing,

$$a\alpha - b\beta + c \frac{\alpha - \beta}{2\alpha\beta + 1} = 0,$$

an equation which is evidently not altered by the interchange of a, α and b, β . The conditions, in order that each bisector may touch two factors, reduce themselves to the three equations,

$$a\alpha - b\beta + c \frac{\alpha - \beta}{2\alpha\beta + 1} = 0,$$

$$b\beta - c\gamma + a \frac{\beta - \gamma}{2\beta\gamma + 1} = 0,$$

$$c\gamma - a\alpha + b \frac{\gamma - \alpha}{2\gamma\alpha + 1} = 0,$$

which are satisfied by the values above found for the quantities a, b, c . The possibility and truth of the geometrical construction are thus demonstrated.

§ 4.

Let it be in the first instance proposed to find the equation of a section touching all or any of the sections $x = 0, y = 0, z = 0$ of the surface of the second order,

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy + pw^2 = 0.$$

Any section whatever of this surface may be written in the form

$$\begin{aligned} & \sqrt{(a\lambda^2 + b\mu^2 + c\nu^2 + 2f\mu\nu + 2g\nu\lambda + 2h\lambda\mu - K)} \rho / \sqrt{-p} \cdot w \\ & + (a\lambda + h\mu + g\nu)x + (h\lambda + b\mu + f\nu)y + (g\lambda + f\mu + c\nu)z = 0, \end{aligned}$$

where

$$\nabla^2 = a\lambda^2 + b\mu^2 + c\nu^2 + 2f\mu\nu + 2g\nu\lambda + 2h\lambda\mu - K,$$

[Book p. 64.] and λ, μ, ν are indeterminate. And considering any other section represented by a like equation,

$$\begin{aligned} & \sqrt{(a\lambda'^2 + b\mu'^2 + c\nu'^2 + 2f\mu'\nu' + 2g\nu'\lambda' + 2h\lambda'\mu' - K)} \rho' / \sqrt{-p} \cdot w \\ & + (a\lambda' + h\mu' + g\nu')x + (h\lambda' + b\mu' + f\nu')y + (g\lambda' + f\mu' + c\nu')z = 0, \end{aligned}$$

where

$$\nabla'^2 = a\lambda'^2 + b\mu'^2 + c\nu'^2 + 2f\mu'\nu' + 2g\nu'\lambda' + 2h\lambda'\mu' - K,$$

it may be shown, by means of the lemma previously given, that the condition of contact is

$$a\lambda\lambda' + b\mu\mu' + c\nu\nu' + f(\mu\nu' + \mu'\nu) + g(\nu\lambda' + \nu'\lambda) + h(\lambda\mu' + \lambda'\mu) + K \frac{\rho\rho'}{p} = \nabla\nabla'.$$

Suppose that λ', μ', ν' satisfy the equations

$$a\lambda' + h\mu' + g\nu' = 0,$$

$$h\lambda' + b\mu' + f\nu' = 0,$$

$$g\lambda' + f\mu' + c\nu' = 0,$$

so that the last-mentioned section becomes $x = 0$; and observing that the first of the above equations may be transformed into

$$a\lambda'^2 + b\mu'^2 + c\nu'^2 = \frac{K}{a},$$

it is easy to obtain $\lambda' = \sqrt{\mathfrak{A}}$, $\mu' = \frac{\mathfrak{H}}{\sqrt{\mathfrak{A}}}$, $\nu' = \frac{\mathfrak{G}}{\sqrt{\mathfrak{A}}}$. The condition of contact thus becomes

$$\frac{K}{\sqrt{\mathfrak{A}}} \rho^2 + K = 0;$$

and taking the under sign, $\lambda = \sqrt{\mathfrak{A}}$, so that if in the above-written equation we establish all or any of the equations $\lambda = \sqrt{\mathfrak{A}}$, $\mu = \sqrt{\mathfrak{B}}$, $\nu = \sqrt{\mathfrak{C}}$, we have the equation of a section touching all or the corresponding sections of the sections

$$x = 0, \quad y = 0, \quad z = 0.$$

In particular we have for a solution of the problem of tactions, the following equation of the section touching $x = 0$, $y = 0$, $z = 0$:

$$(a\sqrt{\mathfrak{A}} + h\sqrt{\mathfrak{B}} + g\sqrt{\mathfrak{C}})x + (h\sqrt{\mathfrak{A}} + b\sqrt{\mathfrak{B}} + f\sqrt{\mathfrak{C}})y + (g\sqrt{\mathfrak{A}} + f\sqrt{\mathfrak{B}} + c\sqrt{\mathfrak{C}})z \\ + \frac{\rho}{\sqrt{-p}} \sqrt{2(\sqrt{\mathfrak{B}\mathfrak{C}} - \mathfrak{F})(\sqrt{\mathfrak{C}\mathfrak{A}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})} \cdot w = 0.$$

Anticipating the use of a notation the value of which will subsequently appear, or putting

[Book p. 65.]

$$K^2 = -f^4 - g^4 - h^4 + 2g^2h^2 + 2h^2f^2 + 2f^2g^2 - \frac{4f^2g^2h^2}{J^2},$$

the equation of the section in question is

$$\frac{f^2}{\sqrt{\mathfrak{A}}}(-f^2 + g^2 + h^2)x + \frac{g^2}{\sqrt{\mathfrak{B}}} (f^2 - g^2 + h^2)y + \frac{h^2}{\sqrt{\mathfrak{C}}} (f^2 + g^2 - h^2)z + \frac{4f^2g^2h^2}{J^2} \frac{\rho\sqrt{-pK}}{\sqrt{\mathfrak{A}\mathfrak{B}\mathfrak{C}}} w = 0.$$

I proceed to investigate a transformation of the equation for the section with an indeterminate parameter λ , which touches the two sections $y = 0$, $z = 0$. We have

$$a\nabla^2 = (a\lambda + h\mu + g\nu)^2 + (b\mu^2 + c\nu^2 - 2f\mu\nu) - \mathfrak{B} - K,$$

or, putting for μ^2 and ν^2 their values $\mathfrak{B}$, $\mathfrak{C}$ in the second term,

$$a\nabla^2 = (a\lambda + h\mu + g\nu)^2 + (\sqrt{\mathfrak{B}\mathfrak{C}} - \mathfrak{F})^2;$$

and introducing instead of λ an indeterminate quantity X , such that

$$a\lambda + h\mu + g\nu = (\sqrt{\mathfrak{B}\mathfrak{C}} - \mathfrak{F})X,$$

we have, also introducing throughout X instead of λ , and completing the substitution of $\sqrt{\mathfrak{B}}$, $\sqrt{\mathfrak{C}}$ for μ , ν , the equation of the section touching $y = 0$, $z = 0$, becomes

$$(ax + hy + gz)X + y\sqrt{\mathfrak{C}} + z\sqrt{\mathfrak{B}} + w\sqrt{-ap}\sqrt{1 + X^2} = 0.$$

It may be remarked here in passing, that this is a very convenient form for the demonstration of the theorem: "If two sections of a surface of the second order touch each other, and are also

tangent sections (of the same class) to two fixed sections, then considering the planes through the axis of the fixed sections and the poles of the tangent sections, and also the tangent planes through this axis, the anharmonic ratio of the four planes is independent of the position of the moveable tangent sections;" where by the axis of the fixed sections is to be understood the line joining their poles.

The sections touching $z = 0$, $x = 0$ and $x = 0$, $y = 0$ are of course

$$\begin{aligned} z\sqrt{\mathfrak{A}} + x\sqrt{\mathfrak{C}} + (hx + by + fz)Y + w\sqrt{-bp}\sqrt{1 + Y^2} &= 0, \\ x\sqrt{\mathfrak{B}} + y\sqrt{\mathfrak{A}} + (gx + fy + cz)Z + w\sqrt{-cp}\sqrt{1 + Z^2} &= 0. \end{aligned}$$

The conditions of contact of the sections represented by the above-written equations would be perhaps most simply obtained directly from the lemma, but it is proper to deduce it from the formula for contact used in the present memoir. If for shortness

$$\Phi(\pm) = a\lambda'\lambda'' + b\mu'\mu'' + c\nu'\nu'' + f(\mu'\nu'' + \mu''\nu') + g(\nu'\lambda'' + \nu''\lambda') + h(\lambda'\mu'' + \lambda''\mu') \pm K \frac{\rho'\rho''}{p},$$

[Book p. 66.] where the symbol $\Phi(\pm)$ is used in order to mark the essentially different character of the results corresponding to the different values of the ambiguous sign, then

$$\begin{aligned} bc\Phi(-) &= f(h\lambda' + b\mu' + f\nu')(g\lambda'' + f\mu'' + c\nu'') \\ &\quad + (\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(g\lambda'' + f\mu'' + c\nu'') \\ &\quad + (\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})(h\lambda'' + b\mu'' + f\nu'') \\ &\quad + f(-a\sqrt{\mathfrak{B}\mathfrak{C}} + h\sqrt{\mathfrak{C}\mathfrak{A}} + g\sqrt{\mathfrak{A}\mathfrak{B}} - aF - (\mathfrak{G}\mathfrak{H} - aF)) \\ &= \dots \\ &= (\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})\{fYZ + \sqrt{\mathfrak{A}}(Y + Z) + f\}. \end{aligned}$$

That is, $bc\Phi(-) = (\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})\{fYZ + \sqrt{\mathfrak{A}}(Y + Z) + f\}$.

What, however, is really required¹, is the value of $\Phi(+)$; to find this, we have

$$\begin{aligned} bc\Phi(+) &= bc\Phi(-) + 2bcK \\ &= (\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})\{fYZ + \sqrt{\mathfrak{A}}(Y + Z) + f\} + 2bcK - 2f(\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H}), \end{aligned}$$

the second line of which is

$$2(\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H}) \left\{ \frac{bcK - f(\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})}{(\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})} \right\} = 2(\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})\sqrt{\mathfrak{A}}\theta,$$

[Book p. 67.] where

$$\theta = \frac{1}{K} (\sqrt{\mathfrak{A}\mathfrak{B}\mathfrak{C}} + \mathfrak{F}\sqrt{\mathfrak{A}} + \mathfrak{G}\sqrt{\mathfrak{B}} + \mathfrak{H}\sqrt{\mathfrak{C}}),$$

and consequently

$$bc\Phi(+) = (\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})[fYZ + \sqrt{\mathfrak{A}}(Y + Z) + f + 2\theta\sqrt{\mathfrak{A}}],$$

¹It may be shown without difficulty that the $(-)$ sign would imply that the sections touching $z = 0$, $x = 0$, and $x = 0$, $y = 0$ were sections touching $x = 0$ at the same point. By taking the $(-)$ sign in each equation we should have the solution of the problem "to determine three sections of a surface of the second order, the two sections of each pair touching one of three given sections at the same point," which is not without interest; the solution may be completed without any difficulty.

a reduction, which on account of its peculiarity, I have thought right to work out in full.

The condition of contact is

$$\Phi(+) = \nabla' \nabla'' = \frac{1}{\sqrt{bc}} (\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H}) \sqrt{1+Y^2} \sqrt{1+Z^2}.$$

Hence finally, the condition in order that the sections

$$\begin{aligned} z\sqrt{\mathfrak{A}} + (hx + by + fz)Y + x\sqrt{\mathfrak{C}} + w\sqrt{-bp}\sqrt{1+Y^2} &= 0, \\ x\sqrt{\mathfrak{B}} + y\sqrt{\mathfrak{A}} + (gx + fy + cz)Z + w\sqrt{-cp}\sqrt{1+Z^2} &= 0, \end{aligned}$$

(the former of which is a section touching $z = 0$, $x = 0$, and the latter a section touching $x = 0$, $y = 0$) may touch, is

$$fYZ + \sqrt{\mathfrak{A}}(Y + Z) + (f + 2\theta\sqrt{\mathfrak{A}}) - \sqrt{bc}\sqrt{1+Y^2}\sqrt{1+Z^2} = 0.$$

The preceding researches show that the solution of Steiner's extension of Malfatti's problem depends on a system of equations, such as the system mentioned at the commencement of the following section.

§ 5.

Consider the system of equations

$$\begin{aligned} \alpha + \beta(Y + Z) + \gamma YZ + \delta\sqrt{1+Y^2}\sqrt{1+Z^2} &= 0, \\ \alpha' + \beta'(Z + X) + \gamma' ZX + \delta'\sqrt{1+Z^2}\sqrt{1+X^2} &= 0, \\ \alpha'' + \beta''(X + Y) + \gamma'' XY + \delta''\sqrt{1+X^2}\sqrt{1+Y^2} &= 0; \end{aligned}$$

these equations may, it will be seen, be solved by quadratics only, when the coefficients satisfy the relations

$$\frac{\gamma - \alpha}{\gamma^2 - \alpha^2} = \frac{\gamma' - \alpha'}{\gamma'^2 - \alpha'^2} = \frac{\gamma'' - \alpha''}{\gamma''^2 - \alpha''^2}.$$

[Book p. 68.] It may be remarked that these equations are satisfied by

$$\beta = 0, \quad \beta' = 0, \quad \beta'' = 0, \quad \gamma = \delta, \quad \gamma' = \delta', \quad \gamma'' = \delta'',$$

or if we write

$$\frac{\alpha}{\gamma} = -l, \quad \frac{\alpha'}{\gamma'} = -m, \quad \frac{\alpha''}{\gamma''} = -n,$$

the equations become, by a simple reduction,

$$\begin{aligned} Y^2 + Z^2 + 2lYZ &= l^2 - 1, \\ Z^2 + X^2 + 2mZX &= m^2 - 1, \\ X^2 + Y^2 + 2nXY &= n^2 - 1, \end{aligned}$$

which are equivalent to the equations discussed in my paper "On a system of Equations connected with Malfatti's Problem and on another Algebraical System," *Cambridge and Dublin Mathematical Journal*, t. iv. [1849] pp. 270–275, [79]; the solution might have been effected by the direct method, which I shall here adopt, of eliminating any one of the variables between the two equations into which it enters, and combining the result with the third equation.

Writing the second and third equations under the form

$$\begin{aligned} A' + B'X + C'\sqrt{1+X^2} &= 0, \\ A'' + B''X + C''\sqrt{1+X^2} &= 0, \end{aligned}$$

the result of the elimination may be presented in the form

$$A'A'' + B'B'' - C'C'' = \sqrt{A'^2 + B'^2 - C'^2} \cdot \sqrt{A''^2 + B''^2 - C''^2},$$

which is most easily obtained by writing $X = \tan \phi$ and operating with the symbol $\cos^{-1}$; but if the rationalized equations be represented by

$$\lambda' + 2\mu'X + \nu'X^2 = 0, \quad \lambda'' + 2\mu''X + \nu''X^2 = 0,$$

the form

$$(\lambda'\nu'' - \lambda''\nu')^2 = 4(\lambda'\nu' - \mu'^2)(\lambda''\nu'' - \mu''^2) - (\lambda'\nu'' - 2\mu'\mu'')^2 \dots$$

leads easily to the result in question. The values which enter are

$$\begin{aligned} A' &= \alpha' + \beta'Z, & A'' &= \alpha'' + \beta''Y, \\ B' &= \beta' + \gamma'Z, & B'' &= \beta'' + \gamma''Y, \\ C' &= \delta'\sqrt{1+Z^2}, & C'' &= \delta''\sqrt{1+Y^2}; \end{aligned}$$

whence, in the first place, by the equation connecting Y, Z ,

$$C'C'' = -\frac{\delta'\delta''}{\delta} \{\alpha + \beta(Y+Z) + \gamma YZ\}.$$

[Book p. 69.] It is obviously convenient that $A'A'' + B'B''$ should be symmetrical with respect to Y and Z , and this will be the case if

$$\alpha'\beta'' + \beta'\gamma'' = \alpha''\beta' + \beta''\gamma',$$

that is, if

$$\beta'(\gamma'' - \alpha'') = \beta''(\gamma' - \alpha'),$$

or, assuming that the equations are symmetrically related to the system, we have the first set of relations between the coefficients, relations which are satisfied by

$$\alpha = \gamma + 2\phi\beta, \quad \alpha' = \gamma' + 2\phi\beta', \quad \alpha'' = \gamma'' + 2\phi\beta'',$$

and the values of $\alpha, \alpha', \alpha''$ will be considered henceforth as given by these conditions.

We have

$$A'A'' + B'B'' - C'C'' = \alpha'\alpha'' + \beta'\beta'' + (\gamma'\beta'' + \gamma''\beta' + 2\phi\beta'\beta'')(Y+Z) + (\beta'\beta'' + \gamma'\gamma'')YZ + \frac{\delta'\delta''}{\delta} \{\alpha + \beta(Y+Z) + \gamma YZ\}$$

The quantities $A'^2 + B'^2 - C'^2, A''^2 + B''^2 - C''^2$ are quadratic functions of Z and Y respectively, and with proper relations between the coefficients, we may assume

$$(A'^2 + B'^2 - C'^2)(A''^2 + B''^2 - C''^2) = U^2 + k[(\alpha + \beta(Y+Z) + \gamma YZ)^2 - \delta^2(1+Y^2)(1+Z^2)],$$

in which U is a linear function of $Y+Z$ and YZ , and k and δ^2 are constants. The first side must, in the first place, be symmetrical with respect to Y and Z , or

$$\alpha'^2 + \beta'^2 - \delta'^2, \quad (\alpha' + \gamma')\beta', \quad \beta'^2 + \gamma'^2 - \delta'^2$$

must be proportional to

$$\alpha''^2 + \beta''^2 - \delta''^2, \quad (\alpha'' + \gamma'') \beta'', \quad \beta''^2 + \gamma''^2 - \delta''^2.$$

But since $(\alpha' + \gamma')\beta'$, $(\alpha'' + \gamma'')\beta''$ are proportional to $\gamma'^2 - \alpha'^2$, $\gamma''^2 - \alpha''^2$, it is only necessary that $\beta'^2 + \gamma'^2 - \delta'^2$ should be proportional to $\beta''^2 + \gamma''^2 - \delta''^2$, that is, that

$$\frac{\gamma'^2 - \alpha'^2}{\beta'^2 + \gamma'^2 - \delta'^2} = \frac{\gamma''^2 - \alpha''^2}{\beta''^2 + \gamma''^2 - \delta''^2},$$

or since the equations are supposed symmetrically related to the system, we must have the second set of relations between the coefficients. Suppose

$$\frac{\gamma - \alpha}{\beta^2 + \gamma^2 - \delta^2} = \frac{\gamma' - \alpha'}{\beta'^2 + \gamma'^2 - \delta'^2} = \frac{\gamma'' - \alpha''}{\beta''^2 + \gamma''^2 - \delta''^2} = -\frac{s}{\phi'},$$

then since

$$\gamma^2 - \alpha^2 = -4(\gamma + \phi\beta)\phi\beta, \quad \&c.,$$

[Book p. 70.] we have

$$\begin{aligned} \delta^2 &= \beta^2 + \gamma^2 - 4s(\gamma + \phi\beta)\beta, \\ \delta'^2 &= \beta'^2 + \gamma'^2 - 4s(\gamma' + \phi\beta')\beta', \\ \delta''^2 &= \beta''^2 + \gamma''^2 - 4s(\gamma'' + \phi\beta'')\beta'', \end{aligned}$$

and δ , δ' , δ'' will be supposed henceforth to satisfy these equations.

We have next

$$\begin{aligned} A'^2 + B'^2 - C'^2 &= 4(\gamma' + \phi\beta')\beta' s(Z + \frac{\beta'}{\phi} + sZ^2), \\ A''^2 + B''^2 - C''^2 &= 4(\gamma'' + \phi\beta'')\beta'' s(Y + \frac{\beta''}{\phi} + sY^2), \end{aligned}$$

which may be simplified by writing

$$s = \frac{\mu - \phi}{1 + \mu^2}, \quad \frac{1 + \mu^2}{\mu - \phi} = \frac{1 + \nu^2}{\mu - \phi},$$

where μ , ν are to be considered as given functions of s and ϕ . These values give

$$\begin{aligned} A'^2 + B'^2 - C'^2 &= 4(\gamma' + \phi\beta')\beta' s(Z + \mu)(Z + \nu), \\ A''^2 + B''^2 - C''^2 &= 4(\gamma'' + \phi\beta'')\beta'' s(Y + \mu)(Y + \nu). \end{aligned}$$

Hence, putting for simplicity

$$b^2 = 4(\gamma' + \phi\beta')(\gamma'' + \phi\beta'')\beta'\beta'',$$

we have

$$b^2 s^2 (Z + \mu)(Z + \nu)(Y + \mu)(Y + \nu) = U^2 + k[(\alpha + \beta(Y + Z) + \gamma Y Z)^2 - \delta^2(1 + Y^2)(1 + Z^2)];$$

and the two sides have next to be expressed in terms of $Y + Z$ and YZ .

If for symmetry we write

$$\xi = 1, \quad \eta = Y + Z, \quad \zeta = YZ,$$

then

$$b^2 s^2 (\mu^2 \xi + \mu \eta + \zeta)(\nu^2 \xi + \nu \eta + \zeta) = U^2 + k[(\alpha \xi + \beta \eta + \gamma \zeta)^2 - \delta^2 \{(\zeta - \xi)^2 + \eta^2\}],$$

and U is now to be considered as a linear function of ξ, η, ζ .

The condition that the first side of the equation may divide into factors, gives an equation for determining k ; and since the condition is satisfied for $k = \infty$, the equation will be linear, and it is easily seen that the value is $k = \frac{b^2 s^2}{\delta^2} (\mu - \nu)^2$. In fact

$$4(\mu^2 \xi + \mu \eta + \zeta)(\nu^2 \xi + \nu \eta + \zeta) = (\mu^2 \xi + \mu \eta + \nu^2 \xi + \nu \eta + 2\zeta)^2 + (\mu - \nu)^2 [(\zeta - \xi)^2 + \eta^2],$$

hence

[Book p. 71.] and we may assume

$$U = \frac{\mu - \nu}{\delta} \left\{ \left(\Lambda + \frac{1}{\Lambda} \right) (\alpha \xi + \beta \eta + \gamma \zeta) - \delta \left(\Lambda - \frac{1}{\Lambda} \right) (\xi + \zeta) \right\},$$

subject to its being shown that the value of Λ given by the comparison of coefficients is constant. The comparison of coefficients gives

$$2\mu + 2\nu = \frac{\mu - \nu}{\delta} \left\{ \left(\Lambda + \frac{1}{\Lambda} \right) \alpha - \left(\Lambda - \frac{1}{\Lambda} \right) \delta \right\},$$

$$4\xi = \frac{\mu - \nu}{\delta} \left\{ \left(\Lambda + \frac{1}{\Lambda} \right) \gamma - \left(\Lambda - \frac{1}{\Lambda} \right) \delta \right\},$$

$$4(1 - \mu^2) = \frac{\mu - \nu}{\delta} \left(\Lambda + \frac{1}{\Lambda} \right) (\gamma - \alpha),$$

the first and third of which give

$$4(1 - \mu^2) = \frac{\mu - \nu}{\delta} \left(\Lambda + \frac{1}{\Lambda} \right) (\gamma - \alpha),$$

which will be identical with the second, if

$$\frac{2(1 - \mu\nu)}{\mu + \nu} = \frac{\beta}{\gamma - \alpha} = -2\phi,$$

which follows at once from the equation $\nu = \frac{1+\mu\phi}{\mu-\phi}$.

Forming next the two equations

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these will be equivalent to a single equation, if

$$\frac{1}{4}(\mu + \nu)^2 \delta^2 = (\mu + \nu)^2 (\beta^2 + \gamma^2) - 4(\mu + \nu) \beta \gamma - 4(\mu^2 - 1) \beta^2,$$

[Book p. 72.] or, finally, if

$$\delta^2 + \delta^2 = \beta^2 + \gamma^2 - 4s\beta \left[\gamma + \frac{\mu\nu-1}{\mu+\nu} \beta \right] = \beta^2 + \gamma^2 - 4s(\gamma + \phi\beta) \beta,$$

which is in fact the case.

Writing the equations for

$$\Lambda + \frac{1}{\Lambda} = \frac{2\delta}{(\mu-\nu)\beta s} (\gamma - 2s\beta),$$

$$\Lambda - \frac{1}{\Lambda} = \frac{2}{(\mu-\nu)\beta s \delta} (\gamma - 2s\beta - 2\phi\beta),$$

in the form

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and substituting in

$$U = \frac{\mu - \nu}{\delta} \left\{ \left(\Lambda + \frac{1}{\Lambda} \right) (\alpha\xi + \beta\eta + \gamma\zeta) - \delta \left(\Lambda - \frac{1}{\Lambda} \right) (\xi + \zeta) \right\},$$

we have

$$U = \frac{2}{\beta s} \left\{ (-\beta + 2s\gamma + 2\phi\gamma) \xi + (\gamma - 2s\beta) \eta + (-\beta + 2s\gamma + 4\phi\gamma) \zeta \right\} \frac{1}{\delta},$$

and consequently, multiplying by

$$b = 2\sqrt{(\gamma' + \phi\beta')(\gamma'' + \phi\beta'')\beta'\beta''},$$

we have

$$\frac{1}{\delta} \sqrt{(\gamma' + \phi\beta')(\gamma'' + \phi\beta'')\beta'\beta''} \left\{ (-\beta + 2s\gamma + 2\phi\gamma) \xi + (\gamma - 2s\beta) \eta + (-\beta + 2s\gamma + 4\phi\gamma) \zeta \right\},$$

or collecting the different terms which enter into the equation

$$A'A'' + B'B'' - C'C'' = \sqrt{A'^2 + B'^2 - C'^2} \cdot \sqrt{A''^2 + B''^2 - C''^2},$$

the result is

$$\begin{aligned} & \left\{ (\alpha'\alpha'' + \beta'\beta'') \xi + (\gamma'\beta'' + \gamma''\beta' + 2\phi\beta'\beta'') \eta + (\beta'\beta'' + \gamma'\gamma'') \zeta \right\} + \frac{\delta'\delta''}{\delta} (\alpha\xi + \beta\eta + \gamma\zeta) \\ & - \frac{1}{\delta} \sqrt{(\gamma' + 2\phi\beta')(\gamma'' + 2\phi\beta'')\beta'\beta''} \left\{ (-\beta + 2s\gamma + 2\phi\gamma) \xi + (\gamma - 2s\beta) \eta + (-\beta + 2s\gamma + 4\phi\gamma) \zeta \right\} = 0, \end{aligned}$$

which, combined with the first equation written under the form

$$\alpha\xi + \beta\eta + \gamma\zeta + \delta\sqrt{(\zeta - \xi)^2 + \eta^2} = 0,$$

determines the ratios of ξ , η , ζ , that is, the values of $Y + Z$ and YZ .

§ 6.

The system of equations

$$\begin{aligned} (f + 2\theta\sqrt{\mathfrak{A}}) + \sqrt{\mathfrak{A}}(Y + Z) + fYZ - \sqrt{bc}\sqrt{1 + Y^2}\sqrt{1 + Z^2} &= 0, \\ (g + 2\theta\sqrt{\mathfrak{B}}) + \sqrt{\mathfrak{B}}(Z + X) + gZX - \sqrt{ca}\sqrt{1 + Z^2}\sqrt{1 + X^2} &= 0, \\ (h + 2\theta\sqrt{\mathfrak{C}}) + \sqrt{\mathfrak{C}}(X + Y) + hXY - \sqrt{ab}\sqrt{1 + X^2}\sqrt{1 + Y^2} &= 0, \end{aligned}$$

where

$$\theta = \frac{1}{K} (\sqrt{\mathfrak{A}\mathfrak{B}\mathfrak{C}} + \mathfrak{F}\sqrt{\mathfrak{A}} + \mathfrak{G}\sqrt{\mathfrak{B}} + \mathfrak{H}\sqrt{\mathfrak{C}}),$$

on which depends the solution of Steiner's extension of Malfatti's problem, is at once seen to belong to the class of equations treated of in the preceding section, and we have $\phi = 0$, $s = 0$. The equations at the conclusion of the preceding section become

$$\begin{aligned} & \left\{ \sqrt{\mathfrak{B}\mathfrak{C}} + gh + 2\theta(g\sqrt{\mathfrak{C}} + h\sqrt{\mathfrak{B}}) + 4\theta^2\sqrt{\mathfrak{B}\mathfrak{C}} \right\} \xi + \left\{ g\sqrt{\mathfrak{C}} + h\sqrt{\mathfrak{B}} + 2\theta\sqrt{\mathfrak{B}\mathfrak{C}} \right\} \eta + \left\{ \sqrt{\mathfrak{B}\mathfrak{C}} + gh \right\} \zeta \\ & - a \left[(f + 2\theta\sqrt{\mathfrak{A}}) \xi + \sqrt{\mathfrak{A}} \eta + f\zeta \right] - \frac{1}{\sqrt{bc}} \sqrt{(g + \theta\sqrt{\mathfrak{B}})(h + \theta\sqrt{\mathfrak{C}})\sqrt{\mathfrak{B}\mathfrak{C}}} \left[(\sqrt{\mathfrak{A}} - 2\theta f) \xi - f\eta + \sqrt{\mathfrak{A}}\zeta \right] = 0, \\ & \left\{ (f + 2\theta\sqrt{\mathfrak{A}}) \xi + \sqrt{\mathfrak{A}} \eta + f\zeta \right\}^2 - bc \left\{ (\xi - \zeta)^2 + \eta^2 \right\} = 0, \end{aligned}$$

which may also be written

$$\begin{aligned} & (\sqrt{\mathfrak{B}\mathfrak{C}} + \mathfrak{F})(\xi + \zeta) + (-a\sqrt{\mathfrak{A}} + g\sqrt{\mathfrak{B}} + h\sqrt{\mathfrak{C}} + 2\theta\sqrt{\mathfrak{B}\mathfrak{C}})(\eta + 2\theta\xi) \\ & - \frac{1}{\sqrt{bc}} \sqrt{(g + \theta\sqrt{\mathfrak{B}})(h + \theta\sqrt{\mathfrak{C}})\sqrt{\mathfrak{B}\mathfrak{C}}} [(\sqrt{\mathfrak{A}} - 2\theta f)\xi - f\eta + \sqrt{\mathfrak{A}}\zeta] = 0, \\ & \sqrt{\mathfrak{A}} \{f(\xi + \zeta) + (\eta + 2\theta\xi)\}^2 - bc \{(\xi - \zeta)^2 + \eta^2\} = 0. \end{aligned}$$

Hence, observing that

$$\begin{aligned} g + \theta\sqrt{\mathfrak{B}} &= \frac{1}{K} (\sqrt{\mathfrak{B}\mathfrak{C}} + \mathfrak{F})(\sqrt{\mathfrak{A}\mathfrak{B}} + \mathfrak{H}), \\ h + \theta\sqrt{\mathfrak{C}} &= \frac{1}{K} (\sqrt{\mathfrak{B}\mathfrak{C}} + \mathfrak{F})(\sqrt{\mathfrak{A}\mathfrak{C}} + \mathfrak{G}), \\ -a\sqrt{\mathfrak{A}} + h\sqrt{\mathfrak{B}} + g\sqrt{\mathfrak{C}} + 2\theta\sqrt{\mathfrak{B}\mathfrak{C}} &= (\sqrt{\mathfrak{B}\mathfrak{C}} + \mathfrak{F}), \end{aligned}$$

and putting for a moment

$$4\lambda = \sqrt{(\sqrt{\mathfrak{A}\mathfrak{B}} + \mathfrak{H})(\sqrt{\mathfrak{A}\mathfrak{C}} + \mathfrak{G})\sqrt{\mathfrak{B}\mathfrak{C}}},$$

and therefore

$$\sqrt{(g + \theta\sqrt{\mathfrak{B}})(h + \theta\sqrt{\mathfrak{C}})\sqrt{\mathfrak{B}\mathfrak{C}}} = (\sqrt{\mathfrak{B}\mathfrak{C}} + \mathfrak{F})\lambda,$$

[Book p. 74.] the first equation divides by $(\sqrt{\mathfrak{B}\mathfrak{C}} + \mathfrak{F})$, and the result is

$$(\xi + \zeta) + \theta(\eta + 2\theta\xi) - \frac{2\lambda}{\sqrt{bc}} [\sqrt{\mathfrak{A}}(\xi + \zeta) - f(\eta + 2\theta\xi) + \sqrt{\mathfrak{A}}(\eta + 2\theta\xi) - (1 + \theta^2)\xi] = 0.$$

Also, by an easy transformation, the second equation becomes

$$\sqrt{\mathfrak{A}} [(\xi + \zeta) - \{\sqrt{\mathfrak{A}}(\xi + \zeta) - f(\eta + 2\theta\xi)\}^2 - bc \{(\xi + \zeta) - (1 + \theta^2)\xi\}^2 = 0,$$

or putting

$$\xi + \zeta + \theta(\eta + 2\theta\xi) = \Theta, \quad \frac{1}{\sqrt{bc}} [\sqrt{\mathfrak{A}}(\xi + \zeta) - f(\eta + 2\theta\xi)] = \Phi, \quad \zeta = \Psi,$$

the equations become

$$\Theta - 2\lambda\Phi = 0, \quad -\Phi^2 + 4\lambda(\Theta - (1 + \theta^2)\Psi) = 0;$$

hence eliminating Φ ,

$$\left(\frac{\Theta}{2\lambda}\right)^2 = \frac{\Theta^2}{(1 + \theta^2)\Psi} \left(1 - \frac{1 + \theta^2}{4\lambda^2}\right),$$

or, observing that

$$1 + \theta^2 = \frac{1}{K^2} (\sqrt{\mathfrak{B}\mathfrak{C}} + \mathfrak{F})(\sqrt{\mathfrak{C}\mathfrak{A}} + \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} + \mathfrak{H}),$$

and reducing, we obtain a determination of Ψ ; also $\Theta = 2\lambda\Phi$ gives

$$\Phi = \frac{K\Theta}{2\sqrt{(\sqrt{\mathfrak{A}\mathfrak{B}} + \mathfrak{H})(\sqrt{\mathfrak{A}\mathfrak{C}} + \mathfrak{G})\sqrt{\mathfrak{B}\mathfrak{C}}}}.$$

Suppose

$$\sqrt{\mathfrak{B}\mathfrak{C}} + \mathfrak{F} = \mathfrak{a}, \quad \sqrt{\mathfrak{B}\mathfrak{C}} - \mathfrak{F} = \mathfrak{a}_1, \quad \frac{\sqrt{\mathfrak{B}\mathfrak{C}}}{\beta\gamma} = \beta_2\gamma_2,$$

then substituting

$$\xi : \eta : \zeta = \left(\frac{\mathfrak{a}}{\mathfrak{a}_1}\right)^{1/2} \left(1 + \frac{\sqrt{\mathfrak{a}}}{\sqrt{\beta_2\gamma_2}}\right) : \dots$$

[Book p. 75.] that is,

$$\begin{aligned}\xi + \zeta + \theta (\eta + 2\theta\xi) - \frac{\sqrt{2}\sqrt{\mathfrak{a}+\mathfrak{a}_1}}{\sqrt{\beta\gamma_1}} [\sqrt{\mathfrak{A}} (\xi + \zeta) - f (\eta + 2\theta\xi)] &= 0, \\ \xi + \zeta + \theta (\eta + 2\theta\xi) - \frac{4bc (\mathfrak{a}+\mathfrak{a}_1)}{\beta_2\gamma_2} \left(1 - \frac{\sqrt{\mathfrak{a}}}{\sqrt{\mathfrak{a}+\mathfrak{a}_1}}\right) \zeta &= 0;\end{aligned}$$

these may be written

$$L\xi + M\eta + N\zeta = 0, \quad L'\xi + M'\eta + N'\zeta = 0,$$

where

$$\begin{aligned}L &= 1 + 2\theta^2 - \frac{\sqrt{2}\sqrt{\mathfrak{a}+\mathfrak{a}_1}}{\sqrt{\beta\gamma_1}} (\sqrt{\mathfrak{A}} - 2\theta f), \\ M &= \theta + \frac{\sqrt{2}\sqrt{\mathfrak{a}+\mathfrak{a}_1}}{\sqrt{\beta\gamma_1}} f, \\ N &= 1 - \frac{\sqrt{2}\sqrt{\mathfrak{a}+\mathfrak{a}_1}}{\sqrt{\beta\gamma_1}} \sqrt{\mathfrak{A}}, \\ L' &= 1 + 2\theta^2 - \frac{4bc (\mathfrak{a}+\mathfrak{a}_1)}{\beta_2\gamma_2} \left(1 - \frac{\sqrt{\mathfrak{a}}}{\sqrt{\mathfrak{a}+\mathfrak{a}_1}}\right), \quad M' = \theta, \quad N' = 1,\end{aligned}$$

or since ξ, η, ζ are equal to 1, $Y + Z, YZ$ respectively,

$$1 : Y + Z : YZ = MN' - M'N : NL' - N'L : LM' - L'M.$$

Also

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whence

$$\begin{aligned}Y + Z &= -\frac{2\sqrt{2}\sqrt{\mathfrak{a}+\mathfrak{a}_1}\sqrt{\beta_2\gamma_2}}{K} \left(1 - \frac{\sqrt{\mathfrak{a}}}{\sqrt{\beta_2\gamma_2}}\right) \left(1 - \frac{\sqrt{\mathfrak{a}_1}}{\sqrt{\mathfrak{a}+\mathfrak{a}_1}}\right) - 2\theta, \\ YZ &= \frac{2\sqrt{2}\sqrt{\mathfrak{a}+\mathfrak{a}_1}\sqrt{\beta_2\gamma_2}}{K} \left(\theta + \frac{\sqrt{2}\sqrt{\mathfrak{a}+\mathfrak{a}_1}}{\sqrt{\beta\gamma_1}} f\right) \left(1 - \frac{\sqrt{\mathfrak{a}}}{\sqrt{\mathfrak{a}+\mathfrak{a}_1}}\right) - 1;\end{aligned}$$

and by forming the analogous expressions for $Z + X$ and ZX , and $X + Y$ and XY , the values of X, Y, Z may be determined. But the equations in question simplify themselves in a remarkable manner by the notation before alluded to.

[Book p. 76.] Suppose

$$\mathfrak{f} = \sqrt[4]{\mathfrak{A}}\sqrt{\sqrt{\mathfrak{B}\mathfrak{C}} - \mathfrak{F}}, \quad \mathfrak{g} = \sqrt[4]{\mathfrak{B}}\sqrt{\sqrt{\mathfrak{C}\mathfrak{A}} - \mathfrak{G}}, \quad \mathfrak{h} = \sqrt[4]{\mathfrak{C}}\sqrt{\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H}}, \quad J = \sqrt{2}\sqrt[4]{\mathfrak{A}\mathfrak{B}\mathfrak{C}},$$

these values give

$$\begin{aligned}\frac{K\sqrt{\mathfrak{A}}}{\sqrt{\mathfrak{B}\mathfrak{C}}} \mathfrak{a} &= 2\mathfrak{f}^2 \left(1 - \frac{\mathfrak{f}^2}{J^2}\right), \\ \frac{K\sqrt{bc}}{\sqrt{\mathfrak{A}}} &= 2\mathfrak{g}\mathfrak{h}\sqrt{1 - \frac{\mathfrak{g}^2}{J^2}}\sqrt{1 - \frac{\mathfrak{h}^2}{J^2}}, \\ \frac{Kf}{\sqrt{\mathfrak{A}}} &= \mathfrak{f}^2 - \mathfrak{g}^2 - \mathfrak{h}^2 + \frac{2\mathfrak{g}^2\mathfrak{h}^2}{J^2}, \\ K\theta &= -\mathfrak{f}^2 - \mathfrak{g}^2 - \mathfrak{h}^2 + 2J^2, \\ K^2 &= -\mathfrak{f}^4 - \mathfrak{g}^4 - \mathfrak{h}^4 + 2\mathfrak{g}^2\mathfrak{h}^2 + 2\mathfrak{h}^2\mathfrak{f}^2 + 2\mathfrak{f}^2\mathfrak{g}^2 - \frac{4\mathfrak{f}^2\mathfrak{g}^2\mathfrak{h}^2}{J^2}.\end{aligned}$$

Applying these results to the preceding formulae and forming for that purpose the equations

$$\frac{2\sqrt{2}\sqrt{\mathfrak{a}+\mathfrak{a}_1}\sqrt{\beta_2\gamma_2}}{K} \cdot 4\mathfrak{g}\mathfrak{h}\lambda = \frac{J^2}{\mathfrak{f}}, \quad \frac{\sqrt{\mathfrak{a}_1}}{\sqrt{\mathfrak{a}+\mathfrak{a}_1}} = \frac{\mathfrak{f}}{J},$$

$$ghK\theta + \frac{J^2}{\sqrt{\mathfrak{A}}}Kf = (J^2 - gh)(f^2 - (g - h)^2) - 2gh(g - h)^2,$$

we have

$$K(Y + Z) + 2K\theta = 4(J^2 - gh)\left(1 - \frac{f}{J}\right),$$

$$K^2YZ + K^2 = \{(J^2 - gh)(f^2 - (g - h)^2) - 2gh(g - h)^2\}\left(1 - \frac{f}{J}\right);$$

the former of which, combined with the similar equations for $Z + X$ and $X + Y$, gives for X, Y, Z the values to be presently stated, and these values will of course verify the second equation and the corresponding equations for ZX and XY .

Recapitulating the preceding notation, if $x = 0, y = 0, z = 0$ are the equations of the given sections, $w = 0$ the equation of the polar plane of their point of intersection with respect to the surface,

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy + pw^2 = 0,$$

the equation of the surface, $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}$ as usual, and

[Book p. 77.] putting

$$f = \sqrt[4]{\mathfrak{A}}\sqrt{\sqrt{\mathfrak{B}\mathfrak{C}} - \mathfrak{F}}, \quad g = \sqrt[4]{\mathfrak{B}}\sqrt{\sqrt{\mathfrak{C}\mathfrak{A}} - \mathfrak{G}}, \quad h = \sqrt[4]{\mathfrak{C}}\sqrt{\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H}}, \quad J = \sqrt{2}\sqrt[4]{\mathfrak{A}\mathfrak{B}\mathfrak{C}},$$

then the equations of the required sections are

$$(ax + hy + gz)X + y\sqrt{\mathfrak{C}} + z\sqrt{\mathfrak{B}} + w\sqrt{-ap}\sqrt{1 + X^2} = 0,$$

$$x\sqrt{\mathfrak{C}} + (hx + by + fz)Y + z\sqrt{\mathfrak{A}} + w\sqrt{-bp}\sqrt{1 + Y^2} = 0,$$

$$x\sqrt{\mathfrak{B}} + y\sqrt{\mathfrak{A}} + (gx + fy + cz)Z + w\sqrt{-cp}\sqrt{1 + Z^2} = 0,$$

where X, Y, Z are to be determined by the following equations,

$$(f + 2\theta\sqrt{\mathfrak{A}}) + \sqrt{\mathfrak{A}}(Y + Z) + fYZ - \sqrt{bc}\sqrt{1 + Y^2}\sqrt{1 + Z^2} = 0,$$

$$(g + 2\theta\sqrt{\mathfrak{B}}) + \sqrt{\mathfrak{B}}(Z + X) + gZX - \sqrt{ca}\sqrt{1 + Z^2}\sqrt{1 + X^2} = 0,$$

$$(h + 2\theta\sqrt{\mathfrak{C}}) + \sqrt{\mathfrak{C}}(X + Y) + hXY - \sqrt{ab}\sqrt{1 + X^2}\sqrt{1 + Y^2} = 0;$$

and the solution of which, putting

$$f = \sqrt[4]{\mathfrak{A}}\sqrt{\sqrt{\mathfrak{B}\mathfrak{C}} - \mathfrak{F}}, \quad g = \sqrt[4]{\mathfrak{B}}\sqrt{\sqrt{\mathfrak{C}\mathfrak{A}} - \mathfrak{G}}, \quad h = \sqrt[4]{\mathfrak{C}}\sqrt{\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H}}, \quad J = \sqrt{2}\sqrt[4]{\mathfrak{A}\mathfrak{B}\mathfrak{C}},$$

is given by the equations

$$KX = \frac{2fgh}{J} + (-f + g + h)^2 - 2(-f + g + h)J,$$

$$KY = \frac{2fgh}{J} + (f - g + h)^2 - 2(f - g + h)J,$$

$$KZ = \frac{2fgh}{J} + (f + g - h)^2 - 2(f + g - h)J.$$

Instead of the direct but very tedious process by which these values of X, Y, Z have been obtained, we may substitute the following *à posteriori verification*.²

²It is perhaps worth noticing that the value of the quantity λ previously made use of is $\lambda^2 = -\frac{1}{K^2}(f^2 - g^2 - h^2 + (J + f - g - h)J) \dots$

We have

$$\begin{aligned} K^2(1+X^2) &= 4(-f+g+h)^2 J^2(1+\frac{f}{J})(1-\frac{g}{J})(1-\frac{h}{J}), \\ K^2\sqrt{1+Y^2}\sqrt{1+Z^2} &= 4(f^2-(g-h)^2)J^2(1-\frac{f}{J})\sqrt{1-\frac{g^2}{J^2}}\sqrt{1-\frac{h^2}{J^2}}, \\ K^2(1+YZ) &= 4(1-\frac{f}{J})\{(J^2-gh)(f^2-(g-h)^2)-2gh(g-h)^2\}, \\ K^2(Y+Z)-2f^2-2g^2-2h^2+4J^2 &= 4(1-\frac{f}{J})(J^2-gh). \end{aligned}$$

[Book p. 78.] Putting also

$$f^2-g^2-h^2+\frac{2g^2h^2}{J^2} K^2(1+YZ) = 4(1-\frac{f}{J})\{(f^2-(g-h)^2)[(J^2-gh)(f^2-(g-h)^2)-2gh(g-h)^2]-4g^2h^2(g-h)^2(J^2-gh)\}$$

we have, after reductions in which the symmetry of the expressions plays an essential part,

$$4(1-\frac{f}{J})(f^2-(g-h)^2)2ghJ^2(1-\frac{g}{J})(1-\frac{h}{J}),$$

and the values obtained above give also

$$2gh\sqrt{1+Y^2}\sqrt{1+Z^2}\sqrt{1+X^2}K^2,$$

which shows that the relation between Y and Z is verified by the assumed values of these quantities, and the other two equations are of course also verified. The solution of the problem will be rendered more complete if the equations of the required sections and of the auxiliary sections made use of in the geometrical construction are expressed in terms of f, g, h, J .

§ 7.

First, to substitute in the equations of the required sections or resultors. Writing the first equation in the form

$$\frac{K^2}{2\sqrt{\mathfrak{B}\mathfrak{C}}}\left[\sqrt{\mathfrak{B}\mathfrak{C}}ax+(hX+\sqrt{\mathfrak{C}})y+(gX+\sqrt{\mathfrak{B}})z+\sqrt{-ap}\sqrt{1+X^2}w\right]=0,$$

the coefficient of x will be

$$\frac{a}{\sqrt{\mathfrak{B}\mathfrak{C}}}\left\{-\frac{f^2}{J^2}(-f^2+g^2+h^2)K^2\left(1-\frac{f}{J}\right)-ghK^2\left(1-\frac{f}{J}\right)\right\}\cdots,$$

[Book p. 79.] or, as it is convenient to write it,

$$f(-f+g+h)\frac{f}{\sqrt{\mathfrak{A}}}\left(1-\frac{f}{J}\right)x.$$

The coefficient of y is

$$\frac{1}{2\sqrt{\mathfrak{B}}}\left\{[-f^4-g^4-h^4+2g^2h^2+2h^2f^2+2f^2g^2-\frac{4f^2g^2h^2}{J^2}]\cdots\right\},$$

or, after all reductions,

$$(f^2-g^2+h^2)\frac{g}{\sqrt{\mathfrak{B}}}\left(1-\frac{f}{J}\right)y,$$

and similarly the coefficient of z is

$$(f^2+g^2-h^2)\frac{h}{\sqrt{\mathfrak{C}}}\left(1-\frac{f}{J}\right)z,$$

and the coefficient of w is

$$\frac{1}{J} \left(1 + \frac{f}{J} \right) (-f + g + h) \cdot 2\sqrt{J^2 - f^2} \sqrt{1 - \frac{g^2}{J^2}} \sqrt{1 - \frac{h^2}{J^2}} \sqrt{-ap} w.$$

Hence, forming the equation of the resultor in question, and by means of it those of the other resultors, the equations of the resultors are

$$\begin{aligned} & f(-f + g + h) \frac{f}{\sqrt{a}} \left(1 - \frac{f}{J} \right) x + g(f^2 - g^2 + h^2) \frac{g}{\sqrt{b}} \left(1 - \frac{f}{J} \right) y \\ & + h(f^2 + g^2 - h^2) \frac{h}{\sqrt{c}} \left(1 - \frac{f}{J} \right) z + 2\sqrt{J^2 - f^2} \sqrt{1 - \frac{g^2}{J^2}} \sqrt{1 - \frac{h^2}{J^2}} \sqrt{-ap} w = 0, \\ & f(-f^2 + g^2 + h^2) \frac{f}{\sqrt{a}} \left(1 - \frac{g}{J} \right) x + g(f - g + h) \frac{g}{\sqrt{b}} \left(1 - \frac{g}{J} \right) y \\ & + h(f^2 + g^2 - h^2) \frac{h}{\sqrt{c}} \left(1 - \frac{g}{J} \right) z + 2\sqrt{J^2 - g^2} \sqrt{1 - \frac{f^2}{J^2}} \sqrt{1 - \frac{h^2}{J^2}} \sqrt{-bp} w = 0, \\ & f(f^2 - g^2 + h^2) \frac{f}{\sqrt{a}} \left(1 - \frac{h}{J} \right) x + g(f^2 + g^2 - h^2) \frac{g}{\sqrt{b}} \left(1 - \frac{h}{J} \right) y \\ & + h(f + g - h) \frac{h}{\sqrt{c}} \left(1 - \frac{h}{J} \right) z + 2\sqrt{J^2 - h^2} \sqrt{1 - \frac{f^2}{J^2}} \sqrt{1 - \frac{g^2}{J^2}} \sqrt{-cp} w = 0, \end{aligned}$$

values which might be somewhat simplified by writing $\mathfrak{f}$, $\mathfrak{g}$, $\mathfrak{h}$, $\mathfrak{w}$ instead of

$$\frac{f}{\sqrt{a}} \left(1 - \frac{f}{J} \right) x, \quad \frac{g}{\sqrt{b}} \left(1 - \frac{g}{J} \right) y, \quad \frac{h}{\sqrt{c}} \left(1 - \frac{h}{J} \right) z, \quad \sqrt{-p} \sqrt{1 - \frac{f^2}{J^2}} \sqrt{1 - \frac{g^2}{J^2}} \sqrt{1 - \frac{h^2}{J^2}} w;$$

and it may be also remarked that the coefficients as well of these formulae as of those which follow may be elegantly expressed in terms of the parts of a triangle having f , g , h for its sides.

The equations of the separators are found by taking the differences two and two of the equations of the resultors (this requires to be verified *à posteriori*); thus subtracting the third equation from the second, the result contains a constant factor

$$\frac{2}{J} [4f^2g^2h^2 - J(f^2 - (g - h)^2)((g + h)^2 - f^2)],$$

[Book p. 80.] equivalent to (after rejection of factor)

...

Rejecting the factor in question and forming the analogous two equations, the equations of the separators are

$$\begin{aligned} & -\frac{g-h}{f} \cdot \frac{f}{\sqrt{a}} \left(1 - \frac{f}{J} \right) x + \frac{g}{\sqrt{b}} \left(1 - \frac{g}{J} \right) y - \frac{h}{\sqrt{c}} \left(1 - \frac{h}{J} \right) z = 0, \\ & \frac{f}{\sqrt{a}} \left(1 - \frac{f}{J} \right) x - \frac{h-f}{g} \cdot \frac{g}{\sqrt{b}} \left(1 - \frac{g}{J} \right) y + \frac{h}{\sqrt{c}} \left(1 - \frac{h}{J} \right) z = 0, \\ & -\frac{f}{\sqrt{a}} \left(1 - \frac{f}{J} \right) x + \frac{g}{\sqrt{b}} \left(1 - \frac{g}{J} \right) y - \frac{f-g}{h} \cdot \frac{h}{\sqrt{c}} \left(1 - \frac{h}{J} \right) z = 0; \end{aligned}$$

[Book p. 81.] and from the mode of formation of these equations it is obvious that the separators have a line in common.

The equations of the determinators being $x = 0$, $y = 0$, $z = 0$, the equations of the *tactors* are

$$\sqrt{b}z - \sqrt{c}y = 0, \quad \sqrt{c}x - \sqrt{a}z = 0, \quad \sqrt{a}y - \sqrt{b}x = 0;$$

and if $\alpha x + \beta y + \gamma z + \delta w = 0$ be the equation of the tactor touching

$$x = 0, \quad \sqrt{b}x - \sqrt{c}z = 0 \text{ and } \sqrt{a}y - \sqrt{b}x = 0,$$

the conditions of contact are

$$\begin{aligned}\mathfrak{A}(\mathfrak{A}\alpha^2 + \dots + \frac{K}{p}\delta^2) &= (\mathfrak{A}\alpha + \mathfrak{H}\beta + \mathfrak{G}\gamma)^2, \\ 2\sqrt{\mathfrak{A}\mathfrak{B}}(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})(\mathfrak{A}\alpha^2 + \dots + \frac{K}{p}\delta^2) &= \{(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})(\alpha\sqrt{\mathfrak{A}} - \beta\sqrt{\mathfrak{B}}) + \gamma(\mathfrak{G}\sqrt{\mathfrak{B}} - \mathfrak{H}\sqrt{\mathfrak{C}})\}^2, \\ 2\sqrt{\mathfrak{A}\mathfrak{C}}(\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\mathfrak{A}\alpha^2 + \dots + \frac{K}{p}\delta^2) &= \{(\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\alpha\sqrt{\mathfrak{A}} - \gamma\sqrt{\mathfrak{C}}) + \beta(\mathfrak{H}\sqrt{\mathfrak{C}} - \mathfrak{G}\sqrt{\mathfrak{B}})\}^2;\end{aligned}$$

whence

$$\frac{1}{\sqrt{\mathfrak{B}}}\sqrt{2\sqrt{\mathfrak{A}\mathfrak{B}}(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})(\mathfrak{A}\alpha + \mathfrak{H}\beta + \mathfrak{G}\gamma)} = (\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})\sqrt{\mathfrak{A}}\alpha - (\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})\sqrt{\mathfrak{B}}\beta + (\mathfrak{G}\sqrt{\mathfrak{B}} - \mathfrak{H}\sqrt{\mathfrak{C}})\gamma,$$

$$\frac{1}{\sqrt{\mathfrak{C}}}\sqrt{2\sqrt{\mathfrak{A}\mathfrak{C}}(\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\mathfrak{A}\alpha + \mathfrak{H}\beta + \mathfrak{G}\gamma)} = (\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})\sqrt{\mathfrak{A}}\alpha + (\mathfrak{H}\sqrt{\mathfrak{C}} - \mathfrak{G}\sqrt{\mathfrak{B}})\beta - (\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})\sqrt{\mathfrak{C}}\gamma,$$

and

$$a\alpha^2 + b\beta^2 + c\gamma^2 - 2f\beta\gamma - 2g\gamma\alpha - 2h\alpha\beta = 0,$$

and putting for a moment

$$\begin{aligned}\sqrt{\mathfrak{A}}\mu &= \sqrt{\mathfrak{B}\mathfrak{C}} - \mathfrak{F} - \sqrt{2}(\sqrt{\mathfrak{B}\mathfrak{C}} - \mathfrak{F}), \\ \sqrt{\mathfrak{A}}\nu &= \sqrt{\mathfrak{B}\mathfrak{C}} - \mathfrak{F} - \sqrt{2}\sqrt{\mathfrak{A}\mathfrak{B}}(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})\end{aligned}$$

after some reductions, and observing that the ratios only of the quantities $\alpha, \beta, \gamma, \delta$ are material, we obtain

$$\alpha = \frac{K}{\sqrt{\mathfrak{A}}}(K + b\nu + g\mu), \quad \beta = \frac{K}{\sqrt{\mathfrak{B}}}(\dots), \quad \gamma = \frac{K}{\sqrt{\mathfrak{C}}}(\dots), \quad \delta = \frac{K}{\sqrt{\mathfrak{A}}}\sqrt{-(b\nu^2 + c\mu^2 + 2f\mu\nu) \cdot p};$$

[Book p. 82.] and it is easily seen also that the coordinates of the point of contact are

$$x = 0, \quad y = \nu, \quad z = \mu, \quad w = -\frac{\delta\sqrt{\mathfrak{A}}}{pK};$$

also

$$\mu = -\frac{f g}{\sqrt{\mathfrak{B}}}\left(1 - \frac{g}{J}\right), \quad \nu = -\frac{f h}{\sqrt{\mathfrak{C}}}\left(1 - \frac{h}{J}\right).$$

Hence substituting and introducing throughout the quantities f, g, h, J , also forming the analogous equations, the equations of the tactors are

$$\begin{aligned}&\{f^2(-f^2 + g^2 + h^2) + (g + h)J(f^2 - (g - h)^2 - \frac{2g^2h^2}{J^2})\}\frac{f}{\sqrt{\mathfrak{A}}}x \\ &\quad + 2\sqrt{X}gh(1 - \frac{g}{J})(1 - \frac{h}{J})(f^2 - (g - h)^2)\sqrt{-p}w = 0, \\ &\{g^2(f^2 - g^2 + h^2) + (h + f)J(g^2 - (h - f)^2 - \frac{2h^2f^2}{J^2})\}\frac{g}{\sqrt{\mathfrak{B}}}y \\ &\quad + 2\sqrt{Y}hf(1 - \frac{h}{J})(1 - \frac{f}{J})(g^2 - (h - f)^2)\sqrt{-p}w = 0, \\ &\{h^2(f^2 + g^2 - h^2) + (f + g)J(h^2 - (f - g)^2 - \frac{2f^2g^2}{J^2})\}\frac{h}{\sqrt{\mathfrak{C}}}z \\ &\quad + 2\sqrt{Z}fg(1 - \frac{f}{J})(1 - \frac{g}{J})(h^2 - (f - g)^2)\sqrt{-p}w = 0.\end{aligned}$$

[Book p. 83.] It is obvious, from the equations, that each separator passes through the point of contact of a tactor and determinant; it consequently only remains to be shown that each separator touches two tactors. Consider the tactor which has been represented by $\alpha x + \beta y + \gamma z + \delta w = 0$; the unreduced values of the coefficients give

$$\mathfrak{A}\alpha + \mathfrak{H}\beta + \mathfrak{G}\gamma = \frac{K^2}{\sqrt{\mathfrak{A}}}, \quad \mathfrak{H}\alpha + \mathfrak{B}\beta + \mathfrak{F}\gamma = \frac{K^2}{\sqrt{\mathfrak{A}}}(\mathfrak{H} + \nu), \quad \mathfrak{G}\alpha + \mathfrak{F}\beta + \mathfrak{C}\gamma = \frac{K^2}{\sqrt{\mathfrak{A}}}(\mathfrak{G} + \mu),$$

$$\sqrt{\mathfrak{A}\alpha^2 + \dots + \frac{K}{p}\delta^2} = \frac{1}{\sqrt{\mathfrak{A}}}(\mathfrak{A}\alpha + \mathfrak{H}\beta + \mathfrak{G}\gamma) = K^2.$$

Represent for a moment the separator $\ell x + my + nz + sw = 0$. Then putting $\mathfrak{A}\ell^2 + \dots + \frac{K}{p}s^2 = \square^2$, since

$$\mathfrak{A}\ell^2 + \dots + \frac{K}{p}s^2 = K^2 \left\{ \ell \left(1 - \frac{f}{j}\right) - \frac{g}{\sqrt{\mathfrak{B}}}(\mathfrak{H} + \nu) \left(1 - \frac{g}{j}\right) \left(1 - \frac{h}{j}\right) - (f - g) \frac{h}{\sqrt{\mathfrak{C}}} \left(1 - \frac{2g}{j}\right) \left(1 - \frac{h}{j}\right) \right\} = \frac{K^2}{j} \left(-(f - g)^2 + h(f + g) \right)$$

the condition of contact becomes

$$\square = \frac{1}{j} \left\{ -(f - g)^2 + h(f + g) - \frac{2fgh}{j} \right\};$$

or, forming the value of $\square^2$ and substituting,

$$(f^2 - (g - h)^2 - \frac{2g^2h^2}{j^2}) \left(1 - \frac{h}{j}\right) \left(1 - \frac{f}{j}\right) \left(1 - \frac{g}{j}\right) = \frac{1}{j} \left(-(f - g)^2 + h(f + g) - \frac{2fgh}{j} \right)^2,$$

which may be verified without difficulty; and thus the construction for the resultors is shown to be true.

§ 8.

Several of the formulae of the preceding sections of this memoir apply to any number of variables. Consider the surface (i.e. hypersurface)

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy + \dots + pw^2 = 0,$$

and the section (i.e. hypersection)

$$(a\lambda + h\mu + g\nu + \dots)x + (h\lambda + b\mu + f\nu + \dots)y + (g\lambda + f\mu + c\nu + \dots)z + \dots + \sqrt{-p}\nabla w = 0,$$

where

$$\nabla^2 = a\lambda^2 + b\mu^2 + c\nu^2 + 2f\mu\nu + 2g\nu\lambda + 2h\lambda\mu + \dots - K,$$

the condition of contact with any other section represented by a similar equation is

$$a\lambda\lambda' + b\mu\mu' + c\nu\nu' + \dots + f(\mu\nu' + \mu'\nu) + g(\nu\lambda' + \nu'\lambda) + h(\lambda\mu' + \lambda'\mu) + \dots + K \frac{\rho\rho'}{p} = \nabla\nabla',$$

where K is the determinant formed with the coefficients $a, b, c, f, g, h, \dots$. And consequently, by establishing all or any of the equations $\lambda = \sqrt{\mathfrak{A}}, \mu = \sqrt{\mathfrak{B}}, \nu = \sqrt{\mathfrak{C}}, \dots$, we have the condition in order that the section in question may touch all or the corresponding sections of the sections $x = 0, y = 0, z = 0, \dots$

$$\text{Let } n \text{ be the number of the variables } x, y, z, \dots; \text{ then } K^{n-1} = \begin{vmatrix} \mathfrak{A} & \mathfrak{H} & \mathfrak{G} & \dots \\ \mathfrak{H} & \mathfrak{B} & \mathfrak{F} & \dots \\ \mathfrak{G} & \mathfrak{F} & \mathfrak{C} & \dots \\ \vdots & & & \end{vmatrix},$$

[Book p. 84.] also

$$K^{n-1} [(a\lambda + h\mu + g\nu + \dots)x + (h\lambda + b\mu + f\nu + \dots)y + (g\lambda + f\mu + c\nu + \dots)z + \dots]$$

$$= - \begin{vmatrix} x & y & z & \dots \\ \lambda & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} \dots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} \dots \\ \nu & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} \dots \\ \vdots & & & \end{vmatrix},$$

whence also

$$K^{n-2}(\nabla^2 + K) = - \begin{vmatrix} \lambda & \mu & \nu & \cdots \\ \lambda & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} \cdots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} \cdots \\ \nu & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} \cdots \\ \vdots & & & \end{vmatrix} \quad \text{or} \quad K^{n-2}\nabla^2 = - \begin{vmatrix} 1 & \lambda & \mu & \nu \cdots \\ \lambda & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} \cdots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} \cdots \\ \nu & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} \cdots \\ \vdots & & & \end{vmatrix},$$

[Book p. 85.] and the equation of the section in question becomes

$$- \begin{vmatrix} x & y & z & \cdots \\ \lambda & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} \cdots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} \cdots \\ \nu & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} \cdots \\ \vdots & & & \end{vmatrix} + K^{\frac{1}{2}(n-1)}\sqrt{-p} \sqrt{- \begin{vmatrix} 1 & \lambda & \mu & \nu \cdots \\ \lambda & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} \cdots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} \cdots \\ \nu & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} \cdots \\ \vdots & & & \end{vmatrix}} \cdot t = 0,$$

also the condition of contact with the corresponding section is

$$- \begin{vmatrix} \pm 1 & \lambda & \mu & \nu \cdots \\ \lambda' & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} \cdots \\ \mu' & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} \cdots \\ \nu' & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} \cdots \\ \vdots & & & \end{vmatrix} = \sqrt{- \begin{vmatrix} 1 & \lambda & \mu & \nu \cdots \\ \lambda & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} \cdots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} \cdots \\ \nu & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} \cdots \\ \vdots & & & \end{vmatrix}} \sqrt{- \begin{vmatrix} 1 & \lambda' & \mu' & \nu' \cdots \\ \lambda' & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} \cdots \\ \mu' & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} \cdots \\ \nu' & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} \cdots \\ \vdots & & & \end{vmatrix}}.$$

In particular the equation of the sections which touches all the sections $x = 0, y = 0, z = 0, \dots$, is

$$- \begin{vmatrix} x & y & z & \cdots \\ \sqrt{\mathfrak{A}} & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} \cdots \\ \sqrt{\mathfrak{B}} & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} \cdots \\ \sqrt{\mathfrak{C}} & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} \cdots \\ \vdots & & & \end{vmatrix} + K^{\frac{1}{2}(n-1)}\sqrt{-p} \sqrt{- \begin{vmatrix} 1 & \sqrt{\mathfrak{A}} & \sqrt{\mathfrak{B}} & \sqrt{\mathfrak{C}} \cdots \\ \sqrt{\mathfrak{A}} & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} \cdots \\ \sqrt{\mathfrak{B}} & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} \cdots \\ \sqrt{\mathfrak{C}} & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} \cdots \\ \vdots & & & \end{vmatrix}} \cdot t = 0.$$

Again, the equations of the section touching $y = 0, z = 0, \dots$ and the section touching $x = 0, z = 0, \dots$ are

$$- \begin{vmatrix} x & y & z & \cdots \\ \lambda & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} \cdots \\ \sqrt{\mathfrak{B}} & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} \cdots \\ \sqrt{\mathfrak{C}} & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} \cdots \\ \vdots & & & \end{vmatrix} + K^{\frac{1}{2}(n-1)}\sqrt{-p} \sqrt{- \begin{vmatrix} 1 & \lambda & \sqrt{\mathfrak{B}} & \sqrt{\mathfrak{C}} \cdots \\ \lambda & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} \cdots \\ \sqrt{\mathfrak{B}} & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} \cdots \\ \sqrt{\mathfrak{C}} & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} \cdots \\ \vdots & & & \end{vmatrix}} \cdot t = 0,$$

$$- \begin{vmatrix} x & y & z & \cdots \\ \sqrt{\mathfrak{A}} & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} \cdots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} \cdots \\ \sqrt{\mathfrak{C}} & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} \cdots \\ \vdots & & & \end{vmatrix} + K^{\frac{1}{2}(n-1)}\sqrt{-p} \sqrt{- \begin{vmatrix} 1 & \sqrt{\mathfrak{A}} & \mu & \sqrt{\mathfrak{C}} \cdots \\ \sqrt{\mathfrak{A}} & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} \cdots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} \cdots \\ \sqrt{\mathfrak{C}} & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} \cdots \\ \vdots & & & \end{vmatrix}} \cdot t = 0,$$

[Book p. 86.] and the condition of contact of these two sections is

$$- \begin{vmatrix} 1 & \lambda & \sqrt{\mathfrak{B}} & \sqrt{\mathfrak{C}} \dots \\ \sqrt{\mathfrak{A}} & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} \dots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} \dots \\ \sqrt{\mathfrak{C}} & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} \dots \\ \vdots & & & \end{vmatrix} = \sqrt{- \begin{vmatrix} 1 & \lambda & \sqrt{\mathfrak{B}} & \sqrt{\mathfrak{C}} \dots \\ \lambda & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} \dots \\ \sqrt{\mathfrak{B}} & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} \dots \\ \sqrt{\mathfrak{C}} & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} \dots \\ \vdots & & & \end{vmatrix}} \sqrt{- \begin{vmatrix} 1 & \sqrt{\mathfrak{A}} & \mu & \sqrt{\mathfrak{C}} \dots \\ \sqrt{\mathfrak{A}} & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} \dots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} \dots \\ \sqrt{\mathfrak{C}} & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} \dots \\ \vdots & & & \end{vmatrix}}.$$

It would seem from the appearance of these equations that there should be some simpler method of obtaining the solution than the method employed in the previous part of this memoir.

115.

NOTE ON THE PORISM OF THE IN-AND-CIRCUMSCRIBED
POLYGON.

[From the *Philosophical Magazine*, vol. vi. (1853), pp. 99–102.]

The equation of a conic passing through the points of intersection of the conics

$$U = 0, \quad V = 0$$

is of the form

$$wU + V = 0,$$

where w is an arbitrary parameter. Suppose that the conic touches a given line, we have for the determination of w a quadratic equation, the roots of which may be considered as parameters for determining the line in question. Let one of the values of w be considered as equal to a given constant k ; the line is always a tangent to the conic

$$kU + V = 0;$$

and taking $w = p$ for the other value of w , p is a parameter determining the particular tangent, or, what is the same thing, the point of contact of this tangent.

Suppose the tangent meets the conic $U = 0$ (which is of course the conic corresponding to $w = \infty$) in the points P, P' , and let θ, ∞ be the parameters of the point P , and θ', ∞ the parameters of the point P' . It follows from my “Note on the Geometrical representation of the Integral $\int dx \div \sqrt{(x+a)(x+b)(x+c)}$,” [113]³ and from the theory of invariants, that if $\square\xi$ represent the “Discriminant” of $\xi U + V$

[Book p. 88.] (I now use the term discriminant in the same sense in which determinant is sometimes used, viz. the discriminant of a quadratic function $ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy$ or $(a, b, c, f, g, h)(x, y, z)^2$, is the determinant $k = abc - af^2 - bg^2 - ch^2 + 2fgh$), and if

$$\Pi\xi = \int_{\infty}^{\xi} \frac{d\xi}{\sqrt{\square\xi}},$$

then the following theorem is true, viz.

“If $(\theta, \infty), (\theta', \infty)$ are the parameters of the points P, P' in which the conic $U = 0$ is intersected by the tangent, the parameter of which is p , of the conic $kU + V = 0$, then the equations

$$\begin{aligned}\Pi\theta &= \Pi p - \Pi k, \\ \Pi\theta' &= \Pi p + \Pi k,\end{aligned}$$

determine the parameters θ, θ' of the points in question.” And again,

“If the variable parameters θ, θ' are connected by the equation

$$\Pi\theta' - \Pi\theta = 2\Pi k,$$

then the line PP' will be a tangent to the conic $kU + V = 0$.” Whence, also,

³I take the opportunity of correcting an obvious error in the note in question, viz. $a^2 + b^2 + c^2 - 2bc - 2ca - 2ab$ is throughout written instead of (what the expression should be) $a^2 + b^2 + c^2 - 2a^2bc - 2b^2ca - 2c^2ab$. [This correction is made, *ante* p. 55.]

“If the sides of a triangle inscribed in the conic $U = 0$ touch the conics

$$kU + V = 0, \quad k'U + V = 0, \quad k''U + V = 0,$$

then the equation

$$\Pi k + \Pi k' + \Pi k'' = 0$$

must hold good between the parameters k, k', k'' .”

And, conversely, when this equation holds good, there are an infinite number of triangles inscribed in the conic $U = 0$, and the sides of which touch the three conics; and similarly for a polygon of any number of sides.

The algebraical equivalent of the transcendental equation last written down is

$$\begin{vmatrix} 1, & k, & \sqrt{\square k} \\ 1, & k', & \sqrt{\square k'} \\ 1, & k'', & \sqrt{\square k''} \end{vmatrix} = 0;$$

let it be required to find what this becomes when $k = k' = k'' = 0$; we have

$$\sqrt{\square k} = A + Bk + Ck^2 + \dots,$$

[Continuation of Paper 115 lies beyond the present slice; the body extends from book p. 88 onward into the next PDF range.]